another (if someone or something is spinning with respect to absolute space then they are *really* spinning; otherwise they are not), this resolution left many people who mull over these issues unsatisfied. Beyond the intuitive sense that no perspective should be "more right" than any other, and beyond the eminently reasonable proposal of Leibniz that only relative motion between material objects has meaning, the concept of absolute space left many wondering how absolute space can allow us to identify true accelerated motion, as with the bucket, while it cannot provide a way to identify true constant velocity motion. After all, if absolute space really exists, it should provide a benchmark for *all* motion, not just accelerated motion. If absolute space really exists, why doesn't it provide a way of identifying where we are located in an absolute sense, one that need not use our position relative to other material objects as a reference point? And, if absolute space really exists, how come it can affect us (causing our arms to splay if we spin, for example) while we apparently have no way to affect it?

In the centuries since Newton's work, these questions were sometimes debated, but it wasn't until the mid-1800s, when the Austrian physicist and philosopher Ernst Mach came on the scene, that a bold, prescient, and extremely influential new view about space was suggested—a view that, among other things, would in due course have a deep impact on Albert Einstein.

To understand Mach's insight—or, more precisely, one modern reading of ideas often attributed to Mach[2]—let's go back to the bucket for a moment. There is something odd about Newton's argument. The bucket experiment challenges us to explain why the surface of the water is flat in one situation and concave in another. In hunting for explanations, we examined the two situations and realized that the key difference between them was whether or not the water was spinning. Unsurprisingly, we tried to explain the shape of the water's surface by appealing to its state of motion. But here's the thing: before introducing absolute space, Newton focused solely on the bucket as the possible reference for determining the motion of the water and, as we saw, that approach fails. There are other references, however, that we could naturally use to gauge the water's motion, such as the laboratory in which the experiment takes place—its floor, ceiling, and walls. Or if we